ABO BLOOD GROUPS

- Determination of ABO blood groups depends upon the immunological reaction between antigen and antibody.
- Landsteiner found two antigens on the surface of RBCs and named them as A antigen and B antigen.
- These antigens are also called agglutinogens because of their capacity to cause agglutination of RBCs.
- He noticed the corresponding antibodies or agglutinins in the plasma and named them anti-A or α-antibody and anti-B or β-antibody.
- A particular agglutinogen and the corresponding agglutinin cannot be present together.
- If present, it causes clumping of the blood.
- Based on this, Karl Landsteiner classified the blood groups.

Landsteiner law

- If an agglutinogen is present on the red cell membrane of an individual, the corresponding agglutinin must be absent in the plasma and
- If an agglutinogen is absent from the cell membrane of RBCs of an individual, the corresponding agglutinin must be present in the plasma.
- Second part of the law does not apply for Rh and many other blood group systems as there are no naturally occurring agglutinins in these systems

ABO Agglutinogens

- In ABO system there are two antigens, the antigen A and the antigen B.
- A antigen has two sub-types, A and A1.
- The genes for antigens for ABO system are ABO genes located on chromosome 9.

Occurrence:

- These antigens are present on the membrane of red cells
- They are also found in other tissues like salivary gland, testes, lungs, kidney and pancreas.

Time of Appearance:

- ABO antigens are present since fetal life.
- They first appear at sixth week of intrauterine life.
- At birth, the concentration is about one-fifth of concentration in adults.

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- Concentration rises slowly to reach a significant level at puberty and reaches maximum at 15–17 years of life
- Therefore, for collection of blood for blood transfusion, the age of the donor should be minimum of 15 years.

Chemical Nature:

- Blood group antigen A and B are complex oligosaccharides.
- For agglutinogen A, terminal sugar is N-acetylgalactosamine
- For agglutinogen B, terminal sugar is galactose.

Blood Groups Based on Antigens

- Based on the presence or absence of A and B antigens on the red cell membrane individually or in group, four blood groups have been described.
- Group A : Antigen A is present
- Group B : Antigen B is present
- Group AB : Both A and B antigens are present
- Group O : Neither A nor B antigen is present

Role of H Gene and H Antigens

- The expression of A and B antigens are dependent on the presence of H genes.
- Basic precursor substance is first converted to H substance (by transferase) under the influence of H gene.
- H substance is partially converted under the influence of A and B genes (and specific transferase) into A and B antigens.
- Some of the H substance remains unconverted.
- Since O group individuals do not have A and B genes, neither A nor B antigen is formed, and these have only H substance (H remain unchanged).

Bombay Blood Group

- In 1952, Bhende, Bhatia and Deshpande discovered a new blood group called as Bombay blood group (Phenotype Oh).
- These individuals lack the H gene, and therefore the basic precursor substance cannot be converted into H substance.
- This in turn results in failure to form A or B antigen.
- Their serum contains anti A and anti B and anti H antibodies.

- These individuals therefore, should be transfused with only Bombay blood group.

ABO Agglutinins

- In ABO system, agglutinins (antibodies) are of two types:
- anti-A (α) and anti-B (β).

Occurrence:

- These antibodies are found in plasma.
- Blood group antibodies are naturally occurring antibodies as they are formed without prior exposure to the antigens

Time and Mechanism of Appearance:

- Anti-A and anti-B agglutinins are not formed during fetal life and are absent at birth.
- They appear in the second week of neonatal life and increase very slowly in concentration to attain the peak level at about 10 years of life.

Chemical Nature:

- The α and β agglutinins are globulins and belong to Ig M category.
- They do not cross the placenta.

Agglutinins against Blood Groups:

- Antibodies are directed against the antigens.
- The individuals with blood group A antigen on the red cell membrane do not have anti-A antibody, rather possess anti-B antibody in their plasma.
- Persons with blood group B do not contain anti-B, rather have anti-A antibody in their plasma.
- Individuals with blood group AB do not have any antibody.
- Individuals with blood group O have both the anti-A and ant-B antibodies in their plasma.

Blood Grouping

- Blood grouping is performed by suspending the red cells of the person's blood group to be checked in the anti-A and anti-B antisera that are commercially available.
- Agglutination occur when the red cells contain that particular antigen.

ABO SYSTEM

Based on the presence or absence of antigen A and antigen B, blood is divided into four groups:

- 'A' group
- 'B' group
- 'AB' group
- 'O' group

Group	Antigen in RBC	Antibody in serum
A	A	Anti-B (β)
B	B	Anti-A (α)
AB	A and B	No antibody
O	No antigen	Anti-A and Anti-B

- Blood having antigen A belongs to 'A' group.
- This blood has β-antibody in the serum.
- Blood with antigen B and α-antibody belongs to 'B' group.
- If both the antigens are present, blood group is called 'AB' group and serum of this group does not contain any antibody.
- If both antigens are absent, the blood group is called 'O' group and both α and β antibodies are present in the serum.

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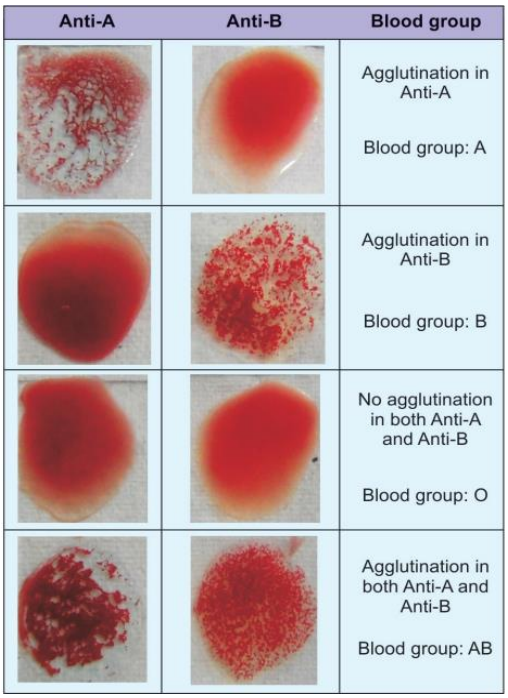
DETERMINATION OF ABO GROUP

Determination of the ABO group is also called blood grouping, blood typing or blood matching

Principle of Blood Typing – Agglutination

- Blood typing is done on the basis of agglutination.
- Agglutination occurs if an antigen is mixed with its corresponding antibody which is called isoagglutinin.
- Agglutination occurs when A antigen is mixed with anti-A or when B antigen is mixed with anti-B.



Rh FACTOR

- Rh factor is an antigen present in RBC.
- This antigen was discovered by Landsteiner and Wiener.
- It was first discovered in Rhesus monkey and hence the name 'Rh factor'.

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Rh Antigen

- There are many Rh antigens but only the D antigen is more antigenic in human.
- The persons having D antigen are called 'Rh positive' and those without D antigen are called 'Rh negative'.

Rh Antibody

- The antibody in the system is called anti-D antibody, which is produced only when an Rh negative individual receives the Rh positive blood
- These antibodies develop very slowly in the first encounter, but form rapidly following subsequent encounters.
- Rh antibody is of Ig G type, which can cross placental barrier.

HEMOLYTIC DISEASE OF FETUS AND NEWBORN – ERYTHROBLASTOSIS FETALIS

- Hemolytic disease is the disease in fetus and newborn, characterized by abnormal hemolysis of RBCs.
- It is due to Rh incompatibility, i.e. the difference between the Rh blood group of the mother and baby.
- Hemolytic disease leads to erythroblastosis fetalis.

Erythroblastosis fetalis

- Erythroblastosis fetalis is a disorder in fetus, characterized by the presence of erythroblasts in blood.
- When a mother is Rh negative and fetus is Rh positive (the Rh factor being inherited from the father), usually the first child escapes the complications of Rh incompatibility.
- This is because the Rh antigen cannot pass from fetal blood into the mother's blood through the placental barrier.
- At the time of parturition (delivery of the child), the Rh antigen from fetal blood may leak into mother's blood because of placental detachment.
- During postpartum period the mother develops Rh antibody in her blood.
- When the mother conceives for the second time and if the fetus is Rh positive again, the Rh antibody from mother's blood crosses placental barrier and enters the fetal blood.
- It is because the Rh antibody is IgG type and can cross placental barrier.

- Rh antibody which enters the fetus causes agglutination of fetal RBCs resulting in hemolysis.
- Severe hemolysis in the fetus causes jaundice.
- To compensate the hemolysis of more and more number of RBCs, there is rapid production of RBCs.
- Many large and immature cells in proerythroblastic stage are released into circulation.
- Because of this, the disease is called erythroblastosis fetalis.
- Ultimately due to excessive hemolysis severe complications develops such as:
    - Severe anemia
    - Hydrops fetalis
    - Kernicterus

PHYSIOLOGICAL BASIS OF BLOOD TRANSFUSION

- Transfusion of whole blood or a component of blood is common in medical practice.
- Be careful for the transmissions of infectious diseases like HIV and hepatitis, and other hazards of transfusions, when transfusion is given frequently.

Indications for Transfusion

- Acute blood loss
- Chronic anemia
- Bone marrow failure
- Clotting factor deficiencies
- Preparation for surgery and during surgery
- Burns
- Anemia

Steps followed for Blood Transfusion

I. Selection of appropriate donor
II. ABO and Rh typing
III. Cross matching
IV. Antibody screening of the patient (to detect the presence of clinically significant antibodies)

I. Selection of Appropriate Donor

- Healthy donors free from transfusion-transmissible diseases should be identified for blood transfusion
- The age of the donor should be 18 years or above and should not be above 60 years.
- There should be minimum of 12 g% Hb in female, and 13 g% Hb in male donors.
- Donor must be screened for AIDS, hepatitis, lymphomas, and protozoal organisms like malaria, filaria, etc.

II. ABO and Rh Typing

- The two major blood group systems for transfusion are ABO and Rh systems.
- ABO incompatibility causes immediate reactions and may even kill the persons.
- The Rh incompatibility causes delayed reactions.
- Therefore, ABO and Rh compatibility must be ensured before any transfusion.
- The concept of universal donor and recipient helps in this process.

Universal Donor

- The blood group O negative (O group and Rh –Ve) persons are considered to be the universal donor because their red cells contain no antigens.
- Therefore, their blood cells on transfusion to any individual do not cause agglutination reaction.
- Hence, blood group O –Ve can be given to any individual

Universal Recipient

- Individuals having blood group AB positive (AB group and Rh +Ve) are considered to be the universal recipient because their plasma contains no antibodies.
- Therefore, they can safely receive blood from any one as they do not cause agglutination reaction against any blood cells.

III. Cross-matching

There are two types of cross-matching:

Major Cross-matching

- In major cross-matching, the cells of the donor are directly matched against the plasma of the recipient
- It is important to ensure that antibodies present in the recipient's plasma do not harm the donor's red cells, this cross-matching is called major cross-matching.

Minor Cross-matching

- In minor cross-matching, the donor's plasma is checked against the red cells of the recipient.
- It is called minor cross-matching, because it is not very important to assess the reactivity of donor's agglutinin against recipient's cells.
- This is because; small volume of the donor's plasma is diluted in a large volume of the recipient's plasma.

IV. Antibody Screening of the Patient

- The patient who receives transfusion, his serum should be screened for various antibodies like autoantibodies that may react against the donor's cells and cause hemolysis.

Collection of Blood

- Venipuncture is done to collect blood:

1. Not more than 350 ml of blood is collected from a single donor at a time.
2. From antecubital vein, blood flows freely into a collecting bottle or bag that contains anticoagulant.
3. Disodium hydrogen citrate is used as anticoagulant
4. The usual collection time is 7–10 min.
5. The blood mixes with 50 to 70 ml of anticoagulant consisting of citrate, phosphate and dextrose (CPD).
6. BP and heart rate are monitored after collection, especially before they get up from the supine posture.
7. Normally, some degree of diaphoresis, bradycardia, hypertension and dizziness accompany this bleeding.
8. Donors are advised not to go for work for rest of the day and should also avoid exercise.

9. They are also advised not to donate blood for at least next three months.

Storage of Blood

- Blood is stored in the blood bank at 4°C.
- Stored blood for transfusion should ideally be used within 2 weeks of storage.

Hazards of Blood Transfusion

Mostly, transfusion reactions occur due to human error:

Due to Mismatched Transfusion:

- When an incompatible blood is transfused, the mismatched transfusion reaction occurs immediately.
- The reaction is primarily due to agglutination of donor's red cells followed by their hemolysis.
- This is called acute hemolytic transfusion reactions.

The complications of mismatched transfusion are:

- Shivering and fever (febrile reactions)
- Hemoglobinemia and hemoglobinuria
- Hemolytic jaundice
- Acute renal failure.

Due to Faulty Techniques of Giving Blood

Due to wrong method of transfusion, following complications may arise:

- Thrombophlebitis: This is a common complication in those who receive repeated transfusions.
- Air Embolism: Air enters the venous circulation and gets lodged at the outlet of the right ventricle and blocks the flow of blood to the lungs.
- Death may occur in severe cases.

Due to Massive Transfusion:

- This occurs when more than 10 units of blood are given within 24 hours or when the total blood volume is exchanged within 24 hours.
- This leads to circulatory overload.

- Cardiac arrhythmias and even sometimes cardiac arrest occur due to high potassium level in the stored blood.
- Febrile Reaction: The patient feels cold and may get rigor due to rise in body temperature. This occurs mainly due to presence of pyrogens in the transfusion apparatus.

Allergic Reactions:

- This is less frequent and is characterized by itching, erythema, nausea, vomiting, and in severe cases may cause anaphylactic reactions.

Transmission of Diseases:

- Hepatitis
- Malaria
- AIDS
- Syphilis.

ANEMIA

Anemia is defined as decreased red cell count or hemoglobin content of blood. It is a disorder, characterized by the reduction in:

- Red blood cell (RBC) count
- Hemoglobin content
- Packed cell volume (PVC)

Haemoglobin concentration is less than:

- 13 g/dL in adult males
- 11.5 g/dL in adult females
- 15 g/dL in newborn
- 9.5 g/dL at 3 months of age.

Classification of Anemia

- Morphological Classification
- Etiological Classification

Morphological Classification:

- Normocytic Normochromic anemia - Size (MCV) and color (MCHC) of RBCs are normal. But the number of RBC is less
- Macrocytic Normochromic anemia- RBCs are larger in size with normal color. RBC count is less.
- Macrocytic Hypochromic anemia - RBCs are larger in size. MCHC is less, so the cells are pale (less colored).
- Microcytic Hypochromic anemia - RBCs are smaller in size with less color.

Normochromic Normocytic Anemia

- MCV, MCH, and MCHC are within the normal range.
- Size and hemoglobin concentration of the red cells are normal
- It usually occurs in following conditions:
    - Blood loss anemia
    - Hemolytic anemia
    - Aplastic anemia

Macrocytic Normochromic Anemia

- The MCV is above the normal.
- The size of the cell will be large
- The red cells are usually normochromic
- Megaloblasts are seen in bone marrow examination

It usually occurs in:

- Megaloblastic anemia that occurs due to deficiency of vitamin B12 or folic acid.

Microcytic Hypochromic Anemia

- MCV, MCH, and MCHC are below normal.
- Microcytosis and Hypochromia of red cells in the blood film
- Micronormoblasts are seen in bone marrow examination.

It usually occurs in:

- Iron-deficiency anemia in which there is inadequate iron for the formation of the heme component of the hemoglobin
- Thalassemia in which the formation of the globin component of hemoglobin is defective.

Macrocytic Hypochromic anemia

- RBCs are larger in size.
- MCHC is less, so the cells are pale (less colored)
- Example : Megaloblastic anemia

Etiological Classification:

1. Hemorrhagic anemia
2. Hemolytic anemia
3. Deficiency anemia
4. Aplastic anemia
5. Anemia of chronic diseases.

1.Hemorrhagic anemia

Cause: Occurs due to excess blood loss.

Types:

Acute hemorrhagic anemia - Acute hemorrhage refers to sudden loss of a large quantity of blood as in the case of accident.

It causes reduction in the total blood volume.

Blood examination shows Normocytic, Normochromic or Normocytic hypochromic anemia

Chronic hemorrhagic anemia

It refers to loss of blood by internal or external bleeding, over a long period of time.

Eg:

- Peptic ulcer
- Purpura
- Hemophilia
- Menorrhagia.

2. Hemolytic Anemia

Cause: Anemia due to excessive hemolysis which is not compensated by increased RBC production is called hemolytic anemia.

Hemolytic anemia's are classified into:

1. Hereditary hemolytic anemia
2. Acquired hemolytic anemia

Hereditary Hemolytic anemia's are:

- Sickle cell anemia
- Thalassemia
- Hereditary spherocytosis

Sickle cell anemia

This is a hereditary disorder in which red cells contain an abnormal hemoglobin called Hb S.

- Hb S is the hemoglobin in which glutamic acid is replaced by valine at the 6th position of beta chain.
- In the deoxygenated state, conformational changes induced by Hb S makes the cell more rigid and deformed to take the shape of a sickle.
- Therefore, cells undergo intravascular hemolysis.
- Blood picture shows Normochromic and normocytic.

Thalassemia

- Thalassemia is a genetically determined disease in which the rate of synthesis of one or more types of hemoglobin polypeptide chain is decreased.
- This results in the hemolysis.

Types of thalassemia:

- α thalassemia
- β thalassemia

α-thalassemia

- Anemia of α-thalassemia occurs due to the failure to synthesize α chain.
- It is more hemolytic than dyserythropoietic.

β thalassemia

- Anemia of β thalassemia occurs due to the failure to synthesize β chain
- There is excess α chain production that damages red cell precursor and red cells.

Hereditary spherocytosis

- Heriditary spherocytosis occurs due to the decreased quantity of spectrin on the red cell membrane that results in spherocytic shape of the cell.
- Spherocytes have a decreased surface area to volume ratio and the cells are more rigid
- Therefore, when cells pass through the splenic pulp, they are destroyed.

Features are:

- Anemia
- Jaundice

The blood picture shows:

- Normocytic and normochromic anemia

Acquired haemolytic anaemias are:

- Immunohaemolytic anaemia (due to antibodies against RBCs)
- Haemolytic anaemia due to direct toxic effects (e.g. in malaria, snake venom, toxic effects of drugs and chemicals),
- Haemolytic anaemia in splenomegaly

3. Deficiency Anemia

a. Iron deficiency anemia
b. Megaloblastic anemia
   - ✓ Megaloblastic anemiadue to deficiency of Vit B12
   - ✓ Megaloblastic anemia due to deficiency of folic acid deficiency

a. Iron deficiency anemia

- This is the commonest form of anemia in the developing countries. It usually occurs due to deficiency of iron in the diet.

Causes:

- Inadequate dietary intake of iron
- Increased loss of iron
- Decreased absorption

Blood picture and red cell indices

- Haemoglobin concentration is decreased.
- RBCs are hypochromic and microcytic.
- Red cell indices like MCV, MCH and MCHC are decreased.

b. Megaloblastic anaemia

- Megaloblastic anaemias are characterized by abnormally large cells of erythrocyte series.
- These are caused by defective DNA synthesis due to deficiency of vitamin B12 and/or folic acid (folate).

Megaloblastic anaemia due to vitamin B12 deficiency

These are caused due to the deficiency of Vitamin B12

Causes:

- Inadequate dietary intake
- Malabsorption of vitamin B12 and folic acid

Megaloblastic anaemia due to folic acid deficiency

Causes:

- Inadequate dietary intake

- Malabsorption, e.g. in coeliac disease, tropical sprue and Crohn's disease.
- Increased demand as occurs in: Physiological conditions such as pregnancy, lactation and infancy
- Excess urinary folate loss, e.g. in active liver disease and congestive heart failure.

Blood picture and red cell indices of megaloblastic anaemia

- RBCs are larger in size (macrocytosis) but contain a normal concentration of haemoglobin (normochromia).
- MCV increases
- MCH increases. MCHC usually normal

4. Aplastic anaemia

Cause:

Aplastic anemia is due to the disorder of red bone marrow. Red bone marrow is reduced and replaced by fatty tissues.

Bone marrow disorder occurs in the following conditions:

- ✓ Repeated exposure to X-ray or gamma ray radiation.
- ✓ Presence of bacterial toxins, quinine, gold salts, benzene, radium, etc.
- ✓ Tuberculosis.
- ✓ Viral infections like hepatitis and HIV infections.

5. Anemia due to chronic diseases

- Anemia of chronic diseases is the second common type of anemia.
- It is characterized by short lifespan of RBCs, caused by disturbance in iron metabolism or resistance to erythropoietin action.
- RBCs are normocytic and normochromic.

  - ➢ Rheumatoid Arthritis
  - ➢ Tuberculosis
  - ➢ Chronic renal failure, in which the erythropoietin secretion decreases

JAUNDICE

- Jaundice or icterus is the condition characterized by yellow coloration of the skin, mucous membrane and deeper tissues due to increased bilirubin level in blood.
- The normal serum bilirubin level is 0.2 to 0.8 mg/dL.
- Jaundice occurs when bilirubin level exceeds 2 mg/dL.

Types of Jaundice

- Prehepatic or hemolytic jaundice
- Hepatic or hepatocellular jaundice
- Post hepatic or obstructive jaundice

Prehepatic or Hemolytic Jaundice

- Hemolytic jaundice occurs because of excessive destruction of RBCs resulting in increased blood level of unconjugated bilirubin.
- In this condition, the excretory function of liver is normal.
- Unconjugated bilirubin increases enormously.
- The liver cells cannot excrete that much excess bilirubin rapidly.
- Unconjugated bilirubin is insoluble in water and is not excreted in urine.
- So, it accumulates in the blood resulting in jaundice.

Hepatic or Hepatocellular jaundice

- Hepatic jaundice occurs due to the damage of hepatic cells.
- Because of the damage, unconjugated and conjugated bilirubin is formed.
- The conjugated bilirubin from liver cannot be excreted and it returns to blood.

Posthepatic or Obstructive

- Post hepatic jaundice occurs because of the obstruction of bile flow.
- The bile cannot be excreted into small intestine.
- So, bile salts and bile pigments enter the circulation.
- The blood contains more amount of conjugated bilirubin